
RF PRIOR: Preserving Global-Context Priors for Efficient Instance Segmentation Transfer

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Abstract

We present an efficient transfer-learning framework that re-purposes a *YOLOv12-x* detector for polygon-based instance segmentation. Our key idea is a Receptive-Field Prior: the largest-receptive-field block (P5) of the backbone, pretrained for detection, is kept *fixed* to preserve global object context, while intermediate low-level blocks (P3–P4) are fine-tuned for boundary precision. We formalize this with a block-diagonal Gaussian prior on backbone weights, yielding a *MAP* objective that acts as implicit adaptation. Multi-scale features from P3–P5 are fused in an attentive decoder to predict per-instance polygons. Experiments show strong and stable performance compared with scratch training or naïve tuning strategy. This approach highlights that carefully constrained reuse of high-level detector features—guided by an explicit inductive bias—can yield strong segmentation.

1 Introduction

Inductive bias—architectural constraints that shape the hypothesis space—is a principal driver of generalization in vision models. Classic CNNs hard-code translation equivariance and locality, while recent hybrids interleave convolution and attention to couple fine detail with global scene context [1–4]. Such designs yield pyramidal hierarchies whose high-resolution stages capture boundaries and textures, and whose low-resolution stages aggregate semantics over large receptive fields. These hierarchies transfer well across tasks: detector backbones provide geometry-aware mid-level cues and dense heads refine them into pixel-accurate segmentations [5, 6]. Yet *how* to preserve useful priors during fine-tuning remains open. Full freezing curbs adaptation under domain shift; full fine-tuning expands the search space, slows convergence, and can overwrite global context [7].

Premise. We observe that the largest receptive-field block (P5) of modern detectors—already enhanced with efficient attention in YOLOv12 [8]—encodes stable scene-level structure that is especially valuable for polygonal instance segmentation. We therefore *anchor* global semantics by freezing P5 and adapt only P3–P4, multi-scale decoder that performs a *single* area-restricted fusion at the P5→P4 interface. This early-adaptive, context-aware recipe sharpens boundaries, reduces texture overfitting, and accelerates convergence.

Contributions.

- **Receptive-Field Prior.** We cast transfer as *MAP* with a block-diagonal Gaussian over backbone weights (Sec. 3.2), unifying a δ -prior on P5 (hard freezing) with zero-mean decay on adaptable blocks in a single objective Eq. (2).

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- **Targeted Global-to-Local Fusion.** We introduce a multi-scale decoder that concentrates area-restricted attention once at the P5-to-P4 fusion, while keeping the high-resolution fusion; this focuses long-range cues exactly where mid-level features benefit most.
- **Automatic BBox-to-Polygon Mining for Transfer.** To leverage box-only corpora under background/label shift, we propose a simple mining module that converts detector boxes into polygon pseudo-masks via candidate segmentation, multi-metric ranking, and contour simplification (Sec. 3.5); integrating these pseudo-polygons into our RF-prior pipeline yields further gains in boundary metrics without sacrificing speed.

2 Related Work

2.1 Inductive Bias and Transfer Regularization

Pyramidal backbones encode complementary scales by design; preserving their semantics during fine-tuning is key for generalization. L^2 -SP contracts parameters toward source weights and mitigates catastrophic drift [7, 9], while subnetwork freezing is used to retain global attention patterns in large-vision models (e.g., ViT-R [10]). Our work adopts a *MAP* view in which a block-diagonal prior fixes the top semantic block (P5) via a δ -prior and regularizes adaptable blocks with zero-mean decay, balancing stability and capacity (Sec. 3.2).

2.2 Global Context for Dense Prediction

Non-local operators [2], criss-cross attention [11], and transformer decoders (SETR [12], SegFormer [13]) inject global context but can be costly at high resolution. Area-restricted attention from the YOLOv12 family [8] offers a compute-aware alternative. Placing a single attention site at the stride-32 to 16 fusion is a targeted compromise that preserves long-range cues while keeping the highest-resolution stage lightweight (Sec. 3.3).

2.3 Automatic Box-to-Polygon Priors

Weakly and box-supervised segmentation has long converted coarse boxes into usable mask supervision via proposal mining and regularization, with BoxSup [14] and the “Simple Does It” line [15] as early milestones, and instance-level formulations such as BoxInst [16]. Promptable segmenters like SAM [17] enable mask proposals from box prompts, while vision-language models such as CLIP [18] provide semantic filtering to favor class-consistent candidates. We situate our approach within this literature by using automatically mined polygons—obtained through proposal selection and contour simplification—as additional priors during transfer. Unlike heavy multi-stage pipelines, our integration couples mined polygons with an RF PRIOR and a Attentive Decoder, emphasizing boundary quality while retaining throughput.

3 Methodology

3.1 Preliminaries

Notation. Given $x \in \mathbb{R}^{3 \times H \times W}$, the backbone $B_{\theta_{bb}}$ produces a feature pyramid $\{F_3, F_4, F_5\}$ at strides $\{8, 16, 32\}$. We decompose $\theta_{bb} = [\theta_b; \theta_5]$ and freeze θ_5 to preserve large-receptive-field semantics inherited from detection; θ_b (P3/P4) and the decoder $H_{\theta_{seg}}$ are optimized for segmentation. The head upsamples and fuses $\{F_3, F_4, F_5\}$ to logits $\hat{M} \in \mathbb{R}^{C \times H \times W}$, with masks $M = \sigma(\hat{M})$. *Intuition.* F_5 supplies global layout/category priors, while lower stages sharpen boundaries and local geometry.

3.2 Receptive-Field Prior

Let $\theta_{bb} = [\theta_3; \theta_4; \theta_5]$ denote backbone blocks (P3–P5) producing $F_\ell = B_{\theta_\ell}(F_{\ell-1})$ with receptive-field radii $R_3 < R_4 < R_5$. We freeze P5 to the detector initialization $\theta_{0,5}$ and fine-tune P3/P4 with SGD (with momentum; L2 weight decay). From a *MAP* perspective, this induces a *block-diagonal* prior (Fig. 1-a):

$$p(\theta_{bb}) \propto \exp\left(-\frac{1}{2} \begin{bmatrix} \theta_3 \\ \theta_4 \end{bmatrix}^\top \text{diag}(\xi_3 I, \xi_4 I) \begin{bmatrix} \theta_3 \\ \theta_4 \end{bmatrix}\right) \delta(\theta_5 - \theta_{0,5}), \quad (1)$$

where $\xi_{3,4}$ coincide with the L2 weight-decay coefficients on P3/P4, and the delta factor encodes the hard freeze of P5 at $\theta_{0,5}$.

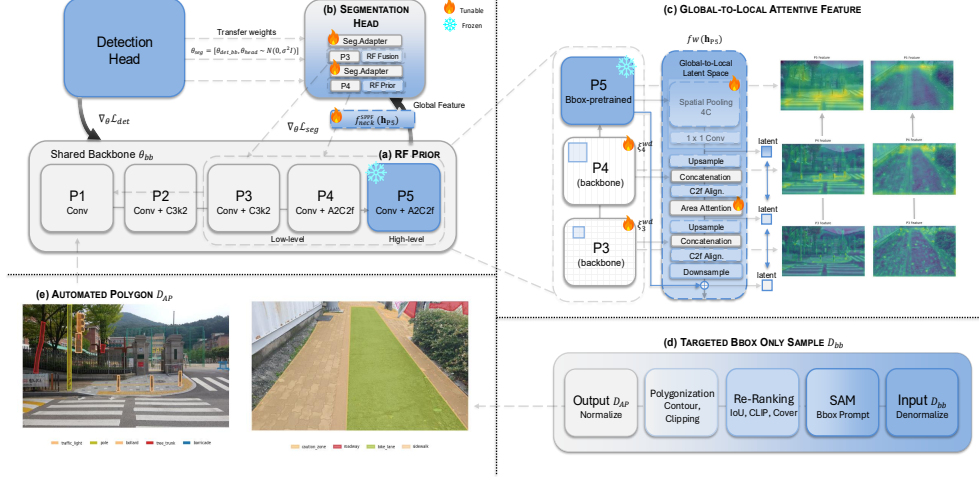


Figure 1: Overview of the proposed transfer-learning scheme. A YOLOv12-x backbone produces multi-scale features P3–P5, where P5 (blue) is attention-augmented with the largest receptive field and is frozen as the RF PRIOR. During segmentation fine-tuning, only P3/P4 (gray) and the segmentation head receive $\nabla \mathcal{L}_{\text{seg}}$ updates. The segmentation head fuses P3, P4, and the fixed P5 context to predict instance masks.

Objective. On $\mathcal{D}_{\text{seg}} = \{(x_i, y_i)\}_{i=1}^N$ we minimize

$$\mathcal{L}_{\text{CC}}(\theta) = \sum_{i=1}^N \underbrace{\ell_{\text{seg}}(H_{\theta_{\text{seg}}} \circ B_{\theta_{\text{bb}}}(x_i), y_i)}_{\text{Unified loss in Sec. 3.6}} + \frac{\xi_3}{2} \|\theta_3\|_2^2 + \frac{\xi_4}{2} \|\theta_4\|_2^2, \quad \text{s.t. } \theta_5 = \theta_{0,5}. \quad (2)$$

Update dynamics. For block $\ell \in \{3, 4\}$ with step size η_ℓ , we employ SGD with momentum coefficient $\mu \in [0, 1)$ and maintain a velocity v_ℓ :

$$v_\ell^{t+1} = \mu v_\ell^t + g_\ell^t + \xi_\ell \theta_\ell^t, \quad (3)$$

$$\theta_\ell^{t+1} = \theta_\ell^t - \eta_\ell v_\ell^{t+1}, \quad g_\ell^t = \nabla_{\theta_\ell} \ell_{\text{seg}}(\theta^t), \quad (4)$$

while the hard constraint yields $\theta_5^{t+1} = \theta_{0,5}$. Writing $\theta_\ell = \theta_{0,\ell} + \Delta_\ell$ (detector init + change) makes the forgetting bias explicit:

$$\Delta_\ell^{t+1} = (1 - \eta_\ell \xi_\ell) \Delta_\ell^t - \eta_\ell (\mu v_\ell^t + g_\ell^t + \xi_\ell \theta_{0,\ell}). \quad (5)$$

Thus, standard L2 weight decay *does not* preserve $\theta_{0,\ell}$; it shrinks both the initialization and the task-driven change toward 0, whereas the P5 freeze preserves global semantics exactly ($\Delta_5 = 0$). (If Nesterov momentum is used, replace g_ℓ^t in the first line with $\nabla_{\theta_\ell} \ell_{\text{seg}}(\theta^t - \eta_\ell \mu v_\ell^t)$; the rest remains analogous.)

Receptive-field locality and polygon adaptation. Backpropagated gradients into block ℓ aggregate supervision from a spatial neighborhood $\mathcal{N}_{R_\ell}(x)$:

$$g_\ell^t \approx \sum_{u \in \mathcal{N}_{R_\ell}(x)} J_\ell^t(x, u)^\top \frac{\partial \ell_{\text{seg}}}{\partial F_\ell^t(u)}, \quad (6)$$

with J_ℓ the feature Jacobian. Since $R_3 < R_4 \ll R_5$ and θ_5 is frozen, updates concentrate on P3/P4 near polygon boundaries where $\partial \ell / \partial F_\ell$ is large. Eq. (5) then yields *compact yet plastic* adjustments in P3/P4 (via the factor $(1 - \eta_\ell \xi_\ell)$).

Context modulation under a frozen P5. Although θ_5 is frozen, the P5 block applies *intra-scale* area attention in an input-conditioned manner. Let

$$\hat{F}_5^{\text{pre}}(x) = B_{\theta_5^{\text{pre}}}^{\text{pre}}(F_4(x)), \quad A(x) = \mathcal{A}_{\text{self}}(\hat{F}_5^{\text{pre}}(x); \theta_A) \in [0, 1]^{H/32 \times W/32},$$

and define the P5 *backbone output* as

$$F_5(x) = A(x) \odot \hat{F}_5(x),$$

with θ_5 and θ_A held fixed during training (no parameter updates). Because both A and $\hat{F}_5$ depend on the input x (and on F_4 , which adapts via $\theta_{3,4}$),

$$\frac{\partial F_5(x)}{\partial x} = \frac{\partial A}{\partial \hat{F}_5} \frac{\partial \hat{F}_5}{\partial x} \odot \hat{F}_5 + A \odot \frac{\partial \hat{F}_5}{\partial x} \neq 0.$$

Moreover, across training steps t , the evolving lower blocks imply

$$F_5^{t+1}(x) - F_5^t(x) \neq 0 \quad \text{even though } \theta_5, \theta_A \text{ remain fixed.}$$

The decoder then consumes (F_5, F_4, F_3) ; cross-scale fusion therein aligns the P5 context with the evolving lower stages and the head.

3.3 Multi-Scale Attentive Decoder

We propose a decoder that adapts the global context encoded by our RF PRIOR to local evidence. This design strengthens cross-scale interaction, enabling more effective fusion of prior-driven global cues with local features (see Figure 1-b,c).

Near-global context via SPPF. A stride-preserving SPPF on F_5 yields $S = [\mathcal{P}_k^{(0)}(F_5) \parallel \mathcal{P}_k^{(1)}(F_5) \parallel \mathcal{P}_k^{(2)}(F_5) \parallel \mathcal{P}_k^{(3)}(F_5)]$, with $\mathcal{P}_k^{(0)} \equiv \text{Id}$ and $\mathcal{P}_k^{(\ell)} = \mathcal{P}_k \circ \mathcal{P}_k^{(\ell-1)}$; $\tilde{F}_5 = \phi_{1 \times 1}(S)$ reprojects to the decoder width. Stacking stride-1 pooling enlarges the effective RF additively, $R_L = 1 + L(k-1)$ (e.g., $k=5, L=3 \Rightarrow R_L=13$ at stride 32).

C2f/A2C2f parametric form. For $X \in \mathbb{R}^{B \times C_{\text{in}} \times H \times W}$,

$$\text{C2f}(X; c, r, s, g, e) = P_{1 \times 1}^{(c)} \left(\text{Cat}[X, \psi^{(1)}(X), \dots, \psi^{(r)}(X)] \right),$$

where each $\psi^{(i)}$ is a bottleneck with expansion e , groups g , and internal shortcut flag $s \in \{0, 1\}$; c is the output width and r the repeat count. Let $U = \text{C2f}(X; c, r, s, g, e)$. Area-restricted attention is then defined by

$$Q = UW_Q, \quad K = UW_K, \quad V = UW_V, \quad W_Q, W_K, W_V \in \mathbb{R}^{C \times d},$$

and a partition $\{\mathcal{A}_r\}_{r=1}^a$ of the spatial grid, with

$$Y = \left(\bigoplus_{r=1}^a \text{softmax}\left(\frac{Q_{\mathcal{A}_r} K_{\mathcal{A}_r}^\top}{\sqrt{d}}\right) V_{\mathcal{A}_r} \right) W_O, \quad W_O \in \mathbb{R}^{d \times C}.$$

We write $A_a(U) = Y$ and define the gated variant

$$\text{A2C2f}(X; c, r, s, g, e, a, \gamma) = U + \gamma A_a(U), \quad \gamma \in \mathbb{R}^C.$$

Top-down fusion with a single attention site. Attention is enabled at the $F_5 \oplus F_4$ fusion; the subsequent high-resolution stage uses C2f. With channel widths and repeats fixed to $(c_4, r_4, s_4) = (512, 2, 0)$ at stride 16 and $(c_3, r_3, s_3) = (256, 2, 0)$ at stride 8, we write

$$\begin{aligned} C_4 &= \text{Cat}[\uparrow_2(\tilde{F}_5), F_4], \quad \hat{C}_4 = P_{1 \times 1}^{(c_4)}(C_4), \quad G_4 = \text{A2C2f}(\hat{C}_4; c_4, r_4, s_4, g=1, e=1, a=4, \gamma), \\ C_3 &= \text{Cat}[\uparrow_2(G_4), F_3], \quad \hat{C}_3 = P_{1 \times 1}^{(c_3)}(C_3), \quad G_3 = \text{C2f}(\hat{C}_3; c_3, r_3, s_3, g=1, e=1). \end{aligned} \quad (7)$$

Complexity. For $Y = A_a(Z)$ with $N=HW$, area partitioning gives $\mathcal{O}(N^2/a)$ time/memory; we use $a=4$ at stride 16 to concentrate attention where global $\rightarrow$ local alignment is most beneficial.

3.4 Global-to-Local Gradient Flow

Let $W_q^{5 \rightarrow 4}$ denote the query projection drawing from F_5 when attending into F_4 . Although θ_5 is frozen, the attention path is trainable and yields

$$\frac{\partial \ell_{\text{seg}}}{\partial F_5} = (W_q^\top (A \odot \frac{\partial \ell_{\text{seg}}}{\partial Z})) R,$$

with attention map A and reshape R . *Interpretation.* The decoder aligns shallow features to the global template in F_5 by implicitly reducing $\mathcal{E} = \sum_k \|F_{5,k} - \phi(F_{4,k})\|^2 + \|F_{5,k} - \phi(F_{3,k})\|^2$, where ϕ is learned projection into the query-aligned space (see Figure 1-c).

3.5 Automatic BBox-to-Polygon Generation

We verify the applicability of our transfer-learning framework to *real-shifted* (background, label) proposed data by polygonizing bbox-only annotations and using them in the transfer stage (Fig. 1-d,e). From a YOLO box $l = (c, \hat{c}_x, \hat{c}_y, \hat{w}, \hat{h})$, we denormalize to $B = [x_1, y_1, x_2, y_2]$ and prompt SAM to obtain candidates $\{M_k\}_{k=1}^K$. For each M_k , we compute $\text{IoU}_k = \frac{|M_k \cap B|}{|M_k \cup B|}$, $\text{Cover}_k = \frac{|M_k \cap B|}{|M_k|}$, and an optional CLIP score $s_k^{\text{clip}} = \cos(f_{\text{img}}(I \odot M_k), f_{\text{ext}}(t_c))$. After per-metric min–max scaling, we rank with $S_k = \alpha \text{IoU}_k + \beta \widetilde{s}_k^{\text{clip}} + \gamma \text{Cover}_k$; if any $\text{IoU}_k \geq \tau$ we take the pixelwise union $\bigvee_{k: \text{IoU}_k \geq \tau} M_k$, else we choose $\arg \max_k S_k$. The selected mask is polygonized via Douglas–Peucker with tolerance ε , vertices are clipped to B , re-normalized by (W, H) , and emitted as $[c, \hat{x}_1, \hat{y}_1, \dots, \hat{x}_n, \hat{y}_n]$.

Design rationale. (1) *Multi-mask union.* SAM can partition a single object; an IoU gate ($\text{IoU} \geq \tau$) with pixelwise union consolidates parts while suppressing off-box regions. (2) *Scoring.* IoU enforces geometric consistency with the box, Cover penalizes off-box leakage, and CLIP (optional) helps disambiguate candidates without altering the given class c . Per-metric min–max normalization calibrates scales so (α, β, γ) are comparable. (3) *Polygonization & clipping.* Douglas–Peucker simplifies boundaries; clipping to B guarantees consistency with the source box; degenerate cases (< 3 vertices) fall back to the rectangle. (4) *Compatibility.* The emitted YOLO-Polygon preserves c and is directly usable in second stage transfer.

3.6 Unified Optimization

We use SGD with momentum and L2 weight decay, together with a short warm-up followed by a linear learning-rate decay (from an initial rate η_0 to a final rate η_{final} over 30 epochs), and a YOLO-style multi-task loss:

$$\mathcal{L} = \lambda_{\text{box}} \text{CIoU} + \lambda_{\text{cls}} \text{BCE} + \lambda_{\text{dfn}} \text{DFL} + \lambda_{\text{mask}} \mathcal{L}_{\text{mask}}, \quad (8)$$

$$\mathcal{L}_{\text{mask}} = \frac{1}{\sum_i s_i^*} \sum_i \frac{s_i^*}{A_i} \sum_{p \in \Omega(b_i^*)} \text{BCE}(\hat{m}_i(p), m_i(p)). \quad (9)$$

s_i^* objectness score, A_i box area, p pixel, b_i^* GT box, $m_i, \hat{m}_i$ GT/pred. masks.

Objectness-/area-normalized mask loss balances small/large instances. Freezing P5 reduces memory (enabling larger batch and stability), while the RF PRIOR (Eq. 2) preserves detector semantics during transfer.

4 Experiments

4.1 Datasets and Experiment Setup

Datasets. Three splits are derived from the SIDEGUIDE traffic-scene corpus [19], all sharing the same 35 object categories. (i) *BBox–DS* is a randomly sampled, down-scaled version of the original bounding-box annotations (max side 512 px); it provides high-level context priors for the detector. (ii) *PolySeg* contains pixel-accurate polygon labels converted to YOLO-style masks. (iii) *SurfaceMask* consists of coarse surface masks for road-layout understanding, integrated with the class indices as *PolySeg* so results are directly comparable for Instance Segmentation. We additionally introduce the *Fixed Objects* dataset and categorize all four datasets along two axes—Background (BG) and Label–Space (LS) shift—relative to *BBox–DS*, our prior source. (1) **Background (BG) shift**—changes in scene/background statistics or capture context; (2) **Label–Space (LS) shift**—changes in the category vocabulary or mask type of label.

PolySeg keeps the same 35 classes as *BBox–DS* but switches from boxes to polygons; the mask–space supervision and per-image region differences induce a *moderate LS* and *mild BG* shift. *SurfaceMask* is polygon-based with

Table 1: Summary of Datasets. We define datasets as Background (BG) and Label–Space (LS) shift.

Datasets (#images)	Custom of SIDEGUIDE & New Dataset				
	Dataset Type	cls	train set	val. set	total
BBox–DS [19]	Prior Source	29	20,097	2,233	22,330
PolySeg [19]	BG shift ▲ , LS shift ▲	29	66,082	7,342	73,424
SurfaceMask [19]	BG shift ✓ , LS shift ✓	6	41,759	4,640	46,399
Fixed Objects (Proposed)	BG shift ▲ , LS shift ✓	15	13,577	1,543	15,120

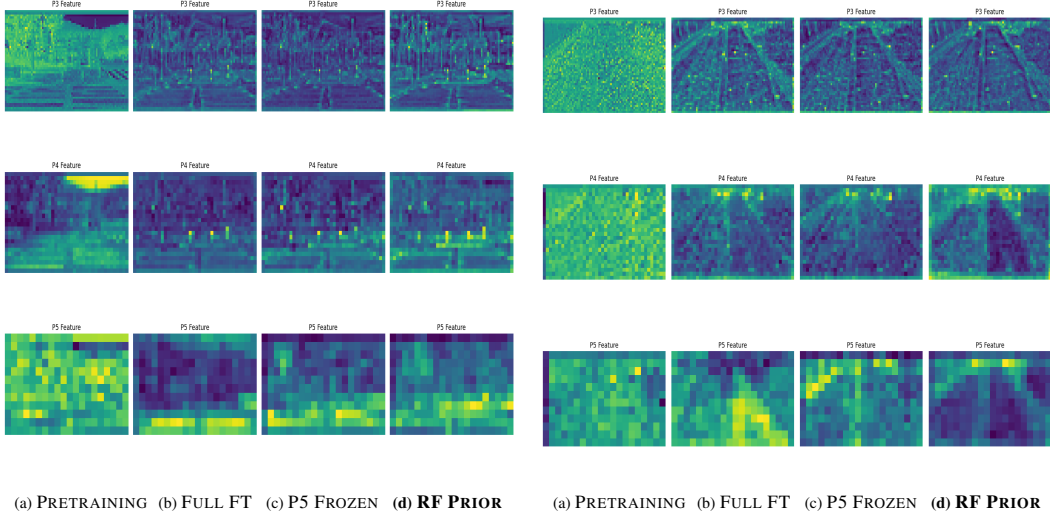


Figure 2: Attention map visualization [20] of **PolysSeg** sample for each model’s trained backbone, θ_{bb} . The mismatch energy $\mathcal{E} = \sum_k \|F_{5,k} - \phi(F_{4/3,k})\|^2$ (Eq. 3.4), is minimized when attention collapses to the frozen P5 context, as visualized.

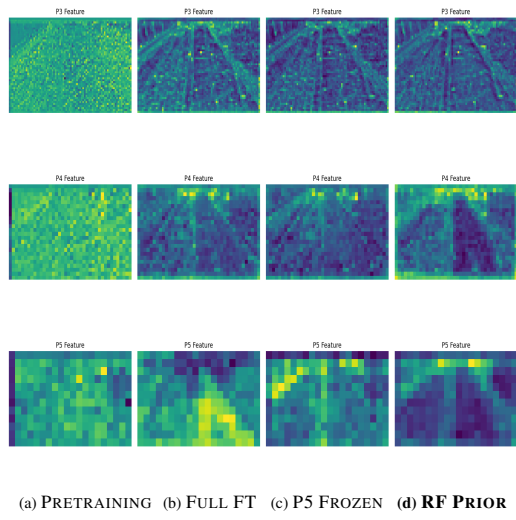


Figure 3: Attention map visualization [20] of **SurfaceMask** for each model’s trained backbone, θ_{bb} . The mismatch energy $\mathcal{E} = \sum_k \|F_{5,k} - \phi(F_{4/3,k})\|^2$ (Eq. 3.4), is minimized when attention collapses to the frozen P5 context, as visualized.

surface-centric labels that differ from *BBox-DS*; although the domain is traffic scenes, images emphasize road surfaces, yielding *strong LS* and *noticeable BG* shift. **Fixed Objects (Proposed)** shares the pedestrian/crosswalk background with *BBox-DS/PolySeg* but uses a different label set focused on fixed infrastructure, thus showing a *clear LS* and *small BG* shift (Table 1 summarizes these relations).

Implementation details. A YOLOv12-x backbone with an information-separated design is used: Models are pre-trained for 500 epochs with image size 640×640 , batch size 8, using an RTX Quadro A6000 (24 GB & fine-tuning for *SIDEGUIDE*). and A100 (40GB & fine-tuning for *Fixed Objects*) in mixed precision. All other hyper-parameters follow Ultralytics defaults.

Evaluation protocol. We report precision, recall, COCO-style mAP_{50} and $mAP_{50:90}$ for boxes, and mask mAP for polygons. All experiments use seed 42. We report the final-epoch score and the per-epoch mean $\pm$ std.dev. to track early-stage adaptation within 30 epochs.

4.2 Ablation Studies and Prior Visualization

Experiments are organized as follows:

- (1) **No Pretraining:** Segmentation training from scratch. (2) **Full Fine-tuning:** No parameters frozen in ℓ_{seg} learning, with transferring pretrained BBox-DS, $\theta_{det.bb}$. (3) **P5 Frozen:** P1–P4 adaptive, P5 fixed in ℓ_{seg} learning, with transferring pretrained BBox-DS, $\theta_{det.bb}$. (4) **Proposed Method (RF PRIOR):** Decoupled backbone, with Multi-Scale Attentive Decoder in $\mathcal{L}_{CC}$. Task is to transfer

Table 2: Performance comparison in Instance Segmentation

Models	#Efficiency $fw(\cdot)$		POLYSEG		SURFACEMASK	
	FPS/GFLOPs	SRI*	mAP_{50}^{val}	$mAP_{50:90}^{val}$	mAP_{50}^{val}	$mAP_{50:90}^{val}$
YOLOv9-E [21]	19.6/1.24	0.58	46.56	27.62	75.59	59.68
YOLOv11-X [22]	36.1/1.60	1.38	46.01	27.41	74.64	59.22
<i>YOLOv12-X Backbone [8]</i>						
NO PRETRAINING*	25.8/1.62	1.00	43.92	26.04	74.73	59.61
FULL FINE-TUNING	25.8/1.62	1.00	44.25	26.22	74.07	59.31
P5 FROZEN	25.9/1.62	1.01	42.95	25.32	74.14	59.58
RF PRIOR	25.2/3.93	2.34	48.42	28.25	<u>75.15</u>	60.15

Table 3: Performance comparison of box (B) and mask (M) metrics on val. set; last-epoch and second-best results.

Models	SIDEGUIDE FOR DETECTION AND INSTANCE SEGMENTATION [19]							
	P (B)	R (B)	mAP_{50}^{val} (B)	$mAP_{50:90}^{val}$ (B)	P (M)	R (M)	mAP_{50}^{val} (M)	$mAP_{50:90}^{val}$ (M)
YOLOv9-E [21]	71.09 _{70.20}	49.85 _{50.09}	55.17 _{54.97}	41.91 _{41.72}	69.69 _{69.49}	46.97 _{46.64}	51.15 _{51.30}	33.25 _{33.15}
YOLOv11-X [22]	66.08 _{68.82}	47.89 _{47.68}	53.93 _{53.47}	40.97 _{40.58}	67.64 _{66.95}	45.39 _{45.59}	50.58 _{50.15}	33.03 _{32.83}
NO PRETRAINING	66.61 _{66.28}	47.15 _{46.65}	52.30 _{52.04}	39.51 _{39.31}	65.41 _{64.72}	44.70 _{44.64}	49.08 _{48.86}	32.08 _{31.95}
└ (50 epochs)	67.95 _{67.62}	49.61 _{49.95}	54.72 _{54.57}	41.84 _{41.71}	69.54 _{69.27}	45.66 _{45.83}	51.25 _{51.12}	33.53 _{33.47}
FULL FINE-TUNING	69.38 _{68.91}	44.97 _{44.88}	51.53 _{51.15}	38.78 _{38.42}	67.72 _{66.85}	43.11 _{43.01}	48.30 _{47.92}	31.48 _{31.22}
P5 FROZEN	63.58 _{64.10}	46.77 _{46.09}	51.61 _{51.26}	38.99 _{38.71}	63.38 _{66.56}	44.04 _{43.48}	48.41 _{48.13}	31.57 _{31.36}
RF PRIOR	70.22 _{69.48}	51.55 _{51.49}	56.95 _{56.63}	42.09 _{41.84}	68.08 _{67.95}	48.91 _{48.08}	52.86 _{52.56}	33.81 _{33.71}
└ (50 epochs)	75.84 _{74.52}	56.69 _{57.08}	63.61 _{63.21}	48.26 _{47.96}	75.11 _{73.96}	52.99 _{52.87}	58.98 _{58.66}	37.87 _{37.64}

Table 4: Performance comparison of box (B) and mask (M) metrics on val. set; mean $\pm$ standard deviation (percentage points) over all epochs.

Models	SIDEGUIDE FOR DETECTION AND INSTANCE SEGMENTATION [19]							
	P (B)	R (B)	mAP_{50}^{val} (B)	$mAP_{50:90}^{val}$ (B)	P (M)	R (M)	mAP_{50}^{val} (M)	$mAP_{50:90}^{val}$ (M)
YOLOv9-E [21]	65.53 ± 7.20	42.09 ± 8.54	46.72 ± 9.98	34.52 ± 8.31	64.59 ± 6.73	40.15 ± 8.04	43.94 ± 9.19	28.05 ± 6.31
YOLOv11-X [22]	64.67 ± 7.05	40.13 ± 7.84	45.08 ± 9.62	33.15 ± 8.03	63.95 ± 6.81	38.24 ± 7.41	42.38 ± 8.93	27.19 ± 6.24
NO PRETRAINING	64.93 ± 6.54	39.01 ± 7.39	43.86 ± 9.21	32.16 ± 7.71	63.88 ± 6.22	37.29 ± 6.99	41.38 ± 8.57	26.51 ± 6.01
└ (50 epochs)	64.43 ± 7.28	39.34 ± 10.85	48.03 ± 14.19	32.08 ± 12.25	65.08 ± 7.17	37.06 ± 9.68	44.45 ± 13.26	25.30 ± 10.60
FULL FINE-TUNING	62.68 ± 7.62	35.71 ± 7.67	40.53 ± 9.71	29.44 ± 8.06	61.59 ± 7.31	34.22 ± 7.34	38.16 ± 9.01	24.34 ± 6.33
P5 FROZEN	63.33 ± 6.14	38.82 ± 7.47	43.11 ± 9.08	31.53 ± 7.57	63.03 ± 6.11	36.98 ± 7.04	40.52 ± 8.42	25.97 ± 5.91
RF PRIOR	64.86 ± 6.44	42.68 ± 8.19	47.60 ± 9.54	34.19 ± 7.80	64.23 ± 5.92	40.16 ± 7.50	44.27 ± 8.71	27.99 ± 6.00
└ (50 epochs)	68.20 ± 6.50	47.13 ± 8.46	52.58 ± 9.66	38.51 ± 8.12	67.16 ± 5.90	44.34 ± 7.81	48.89 ± 8.87	31.08 ± 6.04

the non-overlapping, fine-grained polygon cues provided by SIDEGUIDE—namely *PolySeg* and *SurfaceMask*—and *Fixed Objects* into the *BBox-DS* backbone, so that pedestrian-related objects can be segmented and detected with higher fidelity. Earlier studies on task transfer focused on COCO-scale pre-training, progressive fine-tuning, or freezing all layers before a chosen stage [23, 24]. In contrast, we examine a subtler, label-level shift: how performance changes within a similar domain when annotation granularities and object types differ, rather than when the entire visual domain changes. To this end, we design alignment-aware learning schemes that explicitly account for spatial mis-alignment and category mismatches between the two label spaces.

4.3 Results

Table 2 reports performance for each segmentation sub-task. To show that a small architectural tweak can still transfer well, we introduce the Speed-Retention Index (SRI). We define $SRI_i = (F_i \times G_i) / (F_{base} \times G_{base})$. Our custom head raises GFLOPs (G ; We report GFLOPs scaled by 10^{-2}) by $2.4 \times$ yet reaches an SRI of 2.34, holding FPS (F) loss below 3 % thanks to the memory-bound regime and higher TensorCore utilization. Thus, it improves accuracy while maintaining real-time inference (25 FPS) on an NVIDIA A100 system. RF PRIOR achieves the top mAP on both PolySeg and SurfaceMask. Compared with a 50-epoch, target-focused fine-tune, it raises mAP(50–90), confirming that our Maximum-A-Posteriori formulation boosts accuracy without cross-dataset bottlenecks. Figures 5, 6 illustrate why: in PolySeg, class-specific polygon cues emerge coherently across P3–P5, while in SurfaceMask the model attends to separable polygon lines within the difficult road-surface class over the same feature levels, aligning low- and high-level evidence for stronger segmentation.

4.3.1 Detection and Segmentation on SIDEGUIDE

On SIDEGUIDE (Tables 3–4), all rows except those explicitly marked “(50 epochs)” are trained for 30 epochs. Under this budget, RF PRIOR delivers the strongest last-epoch performance across detection and instance segmentation, reaching mAP_{50}^{val} =**56.95** / **52.86** and $mAP_{50:90}^{val}$ =**42.09** / **33.81**

Table 5: Performance comparison of box (B) and mask (M) metrics on val. set; Top three rows show last-epoch and second-best results. Second rows show mean $\pm$ standard deviation over all epochs.

Models	FIXED OBJECTS FOR DETECTION AND INSTANCE SEGMENTATION							
	P (B)	R (B)	mAP_{50}^{val} (B)	$mAP_{50:90}^{val}$ (B)	P (M)	R (M)	mAP_{50}^{val} (M)	$mAP_{50:90}^{val}$ (M)
Adaptation steps=16.98k, A2C2f scale=1.2 [%]								
NO PRETRAINING	79.77 _{76.98}	64.28 _{62.06}	72.00 _{68.26}	52.54 _{49.32}	69.54 _{66.48}	55.01 _{54.47}	59.14 _{57.76}	40.45 _{38.38}
Γ (w/o AreaAtten.)	79.38 _{78.14}	66.72 _{64.60}	73.44 _{71.72}	53.57 _{50.83}	73.46 _{70.32}	55.78 _{55.60}	62.27 _{60.51}	42.98 _{40.48}
RF PRIOR	80.26 _{75.58}	66.50 _{67.02}	75.62 _{73.07}	55.21 _{53.16}	69.70 _{66.90}	57.23 _{55.56}	63.58 _{60.45}	43.71 _{41.36}
NO PRETRAINING	62.85 $\pm$ 15.94	48.50 $\pm$ 16.10	51.64 $\pm$ 20.84	35.24 $\pm$ 16.37	58.47 $\pm$ 10.26	41.61 $\pm$ 14.21	43.72 $\pm$ 16.73	28.17 $\pm$ 11.96
Γ (w/o AreaAtten.)	61.98 $\pm$ 16.71	49.54 $\pm$ 17.43	52.40 $\pm$ 21.81	34.95 $\pm$ 16.76	57.10 $\pm$ 14.25	42.96 $\pm$ 14.89	44.90 $\pm$ 18.08	28.85 $\pm$ 12.81
RF PRIOR	64.19 $\pm$ 16.06	51.23 $\pm$ 17.20	54.80 $\pm$ 21.51	37.06 $\pm$ 16.85	57.96 $\pm$ 12.99	44.47 $\pm$ 14.35	46.64 $\pm$ 17.38	30.19 $\pm$ 12.60
Adaptation steps=50.94k								
NO PRETRAINING	87.41 _{86.49}	73.83 _{74.14}	83.36 _{83.57}	67.01 _{66.70}	76.53 _{77.66}	63.64 _{64.19}	70.30 _{71.33}	51.23 _{51.48}
Γ (w/o AreaAtten.)	83.72 _{81.40}	81.29 _{81.60}	86.64 _{85.98}	69.55 _{68.67}	73.54 _{70.91}	70.84 _{70.92}	73.51 _{73.15}	53.86 _{53.77}
RF PRIOR	83.99 _{84.24}	80.13 _{79.89}	86.48 _{86.08}	69.84 _{69.38}	72.96 _{72.81}	69.58 _{69.19}	73.05 _{72.64}	53.68 _{53.47}
NO PRETRAINING	77.06 $\pm$ 13.68	63.44 $\pm$ 14.18	70.13 $\pm$ 17.85	52.98 $\pm$ 16.03	68.72 $\pm$ 9.52	54.70 $\pm$ 12.50	59.12 $\pm$ 14.70	41.06 $\pm$ 11.67
Γ (w/o AreaAtten.)	75.88 $\pm$ 13.80	65.84 $\pm$ 15.66	72.03 $\pm$ 18.86	54.17 $\pm$ 17.05	67.55 $\pm$ 11.07	57.09 $\pm$ 13.59	61.31 $\pm$ 15.75	42.75 $\pm$ 12.58
RF PRIOR	76.00 $\pm$ 13.05	66.05 $\pm$ 15.31	72.34 $\pm$ 18.35	54.54 $\pm$ 16.75	67.02 $\pm$ 10.30	57.15 $\pm$ 13.09	61.23 $\pm$ 15.10	42.82 $\pm$ 12.25

for B/M, respectively, which clearly exceeds *No Pretraining* (52.30/49.08 and 39.51/32.08) and *Full Fine-Tuning* (51.53/48.30 and 38.78/31.48). Notably, *P5 Frozen* already matches or slightly surpasses full fine-tuning at the early stage (51.61/48.41 vs. 51.53/48.30 in mAP_{50} for B/M; 38.99/31.57 vs. 38.78/31.48 in $mAP_{50:90}$), supporting the view that preserving large-RF semantics while adapting lower stages accelerates alignment under BG/LS shift. Averaged over epochs (Table 4), RF PRIOR again shows the highest means with moderate variance: for B it attains **47.60** $\pm$ 9.54 (mAP_{50}) and **34.19** $\pm$ 7.80 ($mAP_{50:90}$); for M, **44.27** $\pm$ 8.71 and **27.99** $\pm$ 6.00. Both *No Pretraining* (43.86 $\pm$ 9.21 / 32.16 $\pm$ 7.71 for B; 41.38 $\pm$ 8.57 / 26.51 $\pm$ 6.10 for M) and *Full Fine-Tuning* (40.45 $\pm$ 9.71 / 29.44 $\pm$ 8.06; 38.16 $\pm$ 9.01 / 24.34 $\pm$ 6.33) lag behind. Extending only RF PRIOR to 50 epochs further lifts the means to 52.58 $\pm$ 9.60/38.51 $\pm$ 8.12 (B) and 48.89 $\pm$ 8.87/31.08 $\pm$ 6.04 (M), indicating faster and more stable convergence than training from scratch—even when the latter is given a longer schedule (cf. *No Pretraining* at 50 epochs: 54.72/41.84 for B and 51.25/33.53 for M). Finally, while *YOLOv9-E* exhibits strong precision ($P(B) = 71.09$, $P(M) = 69.69$) consistent with its gradient-concentration design, its attention adaptation under shift remains limited; with the same budget, RF PRIOR attains higher mask mAP_{50} (52.86 vs. 51.15) without relying on additional throughput-oriented tweaks.

5 Discussion

Fixed Objects (Table 5 & Figure 4). Because the FIXED OBJECTS is substantially smaller than SIDEGUIDE, we down-scaled the A2C2f backbone by a factor of 1.2 to avoid overfitting. Even under this tighter capacity budget, RF PRIOR adapts more effectively than the *No Pretraining* and is more stable than RF PRIOR w/o AreaAtten. throughout the early adaptation window.

Early budget (16.98k steps, A2C2f scale=1.2). At the same compute budget shown at the top of Table 5, RF PRIOR achieves $P(B)=80.26_{75.58}$, $R(B)=66.50_{67.02}$, $mAP_{50}^{val}(B)=75.62_{73.07}$, $mAP_{50:90}^{val}(B)=55.21_{53.16}$, and on masks $P(M)=69.70_{66.90}$, $R(M)=57.23_{55.56}$, $mAP_{50}^{val}(M)=63.58_{60.45}$, $mAP_{50:90}^{val}(M)=43.71_{41.36}$. These numbers improve upon *No Pretraining* (e.g., 72.00_{68.26} $mAP_{50}^{val}(B)$ and 59.14_{57.76} $mAP_{50}^{val}(M)$) by +3.62 B- mAP_{50} and +4.44 M- mAP_{50} , and over RF PRIOR w/o AREAATTEN. (73.44_{71.72} / 62.27_{60.51}) by +2.18 (B) and +1.31 (M) absolute mAP_{50} points. Figure 4 corroborates this: both the training and

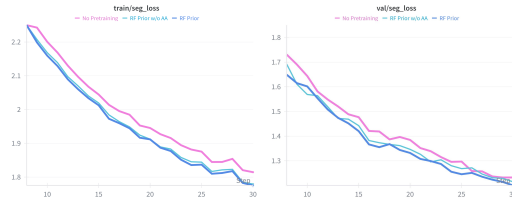


Figure 4: **Left:** Target Adaptation loss (train) and **Right:** Target validation risk in FIXED OBJECTS.

Table 6: Training and validation loss deltas between RF PRIOR and RF PRIOR (w/o AreaAtten.) across 299.6k adaptation steps in SIDEGUIDE. We use $\Delta = \text{RF PRIOR} - \text{RF PRIOR (w/o AreaAtten.)}$. “Last step” is 299.6k. “Global”, “Early” (0–149.8k), and “Late” (149.8k–299.6k) report the mean $\pm$ standard deviation of step-wise deltas over those ranges. “Best step” marks where the most negative Δ occurs. 14,978 steps $\approx$ one full pass over the training set. Additionally, we report the mean log-delta (mean $\Delta_{\log} \pm \text{std}$) and the geometric-mean ratio ($\downarrow\%$) across 0–299.6k steps; positive $\downarrow\%$ indicates average percentage reduction in loss for RF PRIOR.

Loss	Last step	Last Δ	Global mean $\pm$ std	Early mean $\pm$ std	Late mean $\pm$ std	Best step	Best Δ	Mean $\Delta_{\log} \pm \text{std}$	GM ratio ($\downarrow\%$)
train/box_loss	299.6k	-0.10968	-0.25817 ± 0.17441	-0.37392 ± 0.18366	-0.14241 ± 0.02662	15.0k	-0.76653	-0.19280 ± 0.06036	17.54%
train/seg_loss	299.6k	-0.27932	-0.37709 ± 0.08485	-0.44165 ± 0.07239	-0.31253 ± 0.02643	15.0k	-0.60440	-0.17359 ± 0.01635	15.94%
train/cls_loss	299.6k	-0.22018	-0.48829 ± 0.31038	-0.69579 ± 0.32502	-0.28080 ± 0.04546	15.0k	-1.37547	-0.31002 ± 0.06431	26.66%
train/df1_loss	299.6k	-0.19145	-0.36693 ± 0.21945	-0.50646 ± 0.24016	-0.22741 ± 0.02700	15.0k	-0.99273	-0.24601 ± 0.07340	21.81%
val/box_loss	299.6k	-0.03425	-0.23336 ± 0.25246	-0.40081 ± 0.26650	-0.06591 ± 0.03500	15.0k	-0.97990	-0.19155 ± 0.14490	17.43%
val/seg_loss	299.6k	-0.00963	-0.19094 ± 0.20008	-0.34185 ± 0.17992	-0.04003 ± 0.03913	15.0k	-0.61513	-0.09980 ± 0.08731	9.50%
val/cls_loss	299.6k	-0.02323	-0.37914 ± 0.48071	-0.67547 ± 0.53793	-0.08282 ± 0.05776	15.0k	-1.92284	-0.25264 ± 0.20725	22.33%
val/df1_loss	299.6k	-0.08400	-0.24929 ± 0.22282	-0.39044 ± 0.24463	-0.10815 ± 0.02653	15.0k	-0.93564	-0.18350 ± 0.11495	16.76%

validation seg_loss curves for RF PRIOR sit below the others from the earliest steps onward, and the corresponding mAP rises sooner in the adaptation trajectory.

Larger budget (50.94k steps). The gains persist when we triple the adaptation steps (bottom half of Table 5). RF PRIOR reaches about **86.5** B-mAP₅₀ and **73.1** M-mAP₅₀ with strong precision/recall (e.g., $\approx 84/80$ on boxes and $\approx 73/70$ on masks). The second rows in each block report the epoch with the second-best checkpoint, and the bottom rows report the mean $\pm$ std over all checkpoints; in both summaries RF PRIOR remains at least competitive on boxes and consistently better on masks. We attribute the slightly larger variance on FIXED OBJECTS to genuine domain shift: while BBox-DS and outdoor backgrounds overlap with the training distribution, *indoor* backgrounds appear in the validation set only. This out-of-distribution shift inflates variance relative to SIDEGUIDE, yet RF PRIOR still secures higher mAP early and maintains a safe margin over *No Pretraining* (Table 5, Figure 4). Compared to RF PRIOR W/O AREAATTEN., the loss traces show a more consistent gap to *No Pretraining*, suggesting that AreaAtten. dampens oscillations from the indoor/outdoor mixture.

SideGuide (Table 6). We summarize stepwise loss deltas at equal budgets, using $\Delta = \text{RF PRIOR} - \text{RF PRIOR (w/o AreaAtten.)}$. Across the full 0–299.6k steps, RF PRIOR reduces losses relative to its ablation: *train*—box_loss ($\Delta_{\log} = -0.1928 \pm 0.0604$, GM ratio $\downarrow$ **17.54%**), seg_loss (-0.1736 ± 0.0163 , $\downarrow$ **15.94%**), cls_loss (-0.3100 ± 0.0643 , $\downarrow$ **26.66%**), df1_loss (-0.2460 ± 0.0734 , $\downarrow$ **21.81%**); *validation*—box_loss (-0.1916 ± 0.1449 , $\downarrow$ **17.43%**), seg_loss (-0.0998 ± 0.0873 , $\downarrow$ **9.50%**), cls_loss (-0.2526 ± 0.2073 , $\downarrow$ **22.33%**), df1_loss (-0.1835 ± 0.1150 , $\downarrow$ **16.76%**). The most negative deltas (“best step”) consistently occur around **15.0k** steps, indicating that AreaAtten. not only lowers the average risk but also accelerates early adaptation—precisely the regime of interest for rapid deployment.

Takeaways. (i) With a smaller backbone on FIXED OBJECTS, RF PRIOR already yields better mAP at low budgets while keeping losses lower and steadier (Figure 4); (ii) variance is higher than SIDEGUIDE due to indoor backgrounds unique to validation, yet the method preserves its early-step advantage; (iii) on SIDEGUIDE, the quantitative loss analysis shows clear, *systematic* reductions over the ablation across all four losses, confirming that RF PRIOR is both sample-efficient and robust to target shift. Accordingly, we will extend this line of work across a broader spectrum of real-world datasets, toward a more stable and sustainable source–target framework.

6 Conclusion

We proposed a simple, effective strategy for efficient instance segmentation transfer by explicitly preserving global receptive-field priors derived from detection tasks. Our method significantly enhances segmentation performance and training efficiency by freezing the deepest, globally-contextualized block (P5) with aligning decoder. This approach bridges detection and segmentation tasks effectively, presenting a practical transfer learning strategy adaptable to various multi-task vision frameworks.

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A Technical Appendices and Supplementary Material

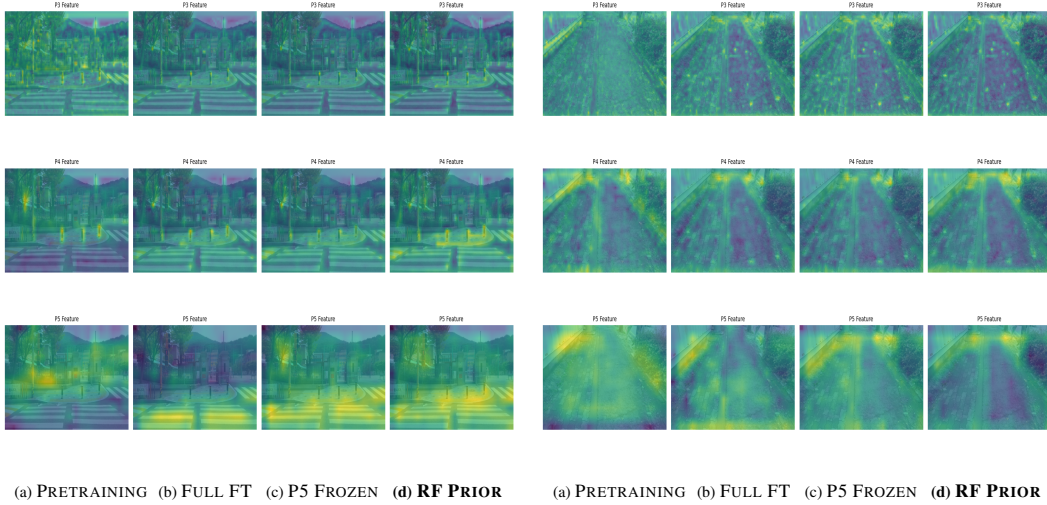


Figure 5: Attention map (overlaid on the original image) visualization [20] of **PolySeg** sample for each model’s trained backbone, θ_{bb} . The mismatch energy $\mathcal{E} = \sum_k \|F_{5,k} - \phi(F_{4/3,k})\|^2$ (Eq. 3.4), is minimized when attention collapses to the frozen P5 context, as visualized.

Figure 6: Attention map (overlaid on the original image) visualization [20] of **SurfaceMask** for each model’s trained backbone, θ_{bb} . The mismatch energy $\mathcal{E} = \sum_k \|F_{5,k} - \phi(F_{4/3,k})\|^2$ (Eq. 3.4), is minimized when attention collapses to the frozen P5 context, as visualized.

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